

Tables



1 Consider the function $f(x) = \frac{1}{7-x}$.

a Complete the first table of values, in which $x < 7$

x	5	6	6.9	6.99
$f(x)$				

b Complete the second table of values, in which $x > 7$

x	9	8	7.1	7.01
$f(x)$				

c Find the limit of $f(x)$ as the value of x approaches 7.

2 Consider the function $f(x) = 5x^2 + 1$:

a Complete the table to find the exact values of $f(x)$ as x gets closer and closer to 2 from the left, and closer and closer to 2 from the right:

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$						

b Find the value of $\lim_{x \rightarrow 2} (5x^2 + 1)$.

3 Consider the function $f(x) = \frac{2-x}{x^2+2}$:

a Complete the table to find the values of $f(x)$ as x gets closer and closer to 0 from the left, and closer and closer to 0 from the right. Round your answers to four decimal places.

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x)$						

b Find the value of $\lim_{x \rightarrow 0} \left(\frac{2-x}{x^2+2} \right)$.

4 Consider the function $f(x) = \frac{x^2 - 4x}{x - 4}$.

a Complete the table to find the values of $f(x)$ as x gets closer and closer to 4 from the left, and closer and closer to 4 from the right:

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$						

b Find the value of $\lim_{x \rightarrow 4} \left(\frac{x^2 - 4x}{x - 4} \right)$.



- 5 Consider the function $f(x) = \frac{x^3 + x + 2}{x + 1}$.

a Complete the following table:

x	-1.1	-1.01	-1.001	-0.999	-0.99	-0.9
$f(x)$						

b Find $\lim_{x \rightarrow -1} f(x)$.

- 6 Consider the function $f(x) = \frac{\sqrt{x} + 4}{x - 5}$:

a Complete the following table, rounding all values to two decimal places:

x	4.9	4.99	4.999	5.001	5.01	5.1
$f(x)$						

b Does $\lim_{x \rightarrow 5} f(x)$ exist? Explain your answer.

- 7 Consider the limit: $\lim_{x \rightarrow -5} \left(\frac{x^2 + 4}{x + 5} \right)$.

a Complete the table.

x	-5.1	-5.01	-5.001	-5	-4.999	-4.99	-4.9
$\frac{x^2 + 4}{x + 5}$				—			

b Does the above limit exist? Explain your answer.

- 8 Consider the limit: $\lim_{x \rightarrow 3} \left(\frac{e^{x-3} + x - 4}{x - 3} \right)$.

a Complete the following table:

x	2.9	2.99	2.999	3	3.001	3.01	3.1
$\left(\frac{e^{x-3} + x - 4}{x - 3} \right)$				—			

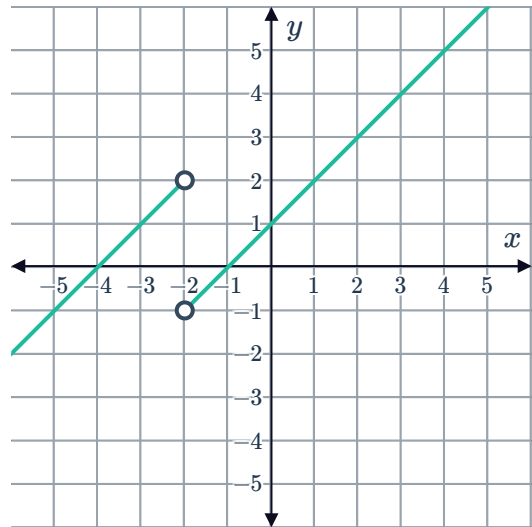
b Does the above limit exist? Explain your answer.

Graphs



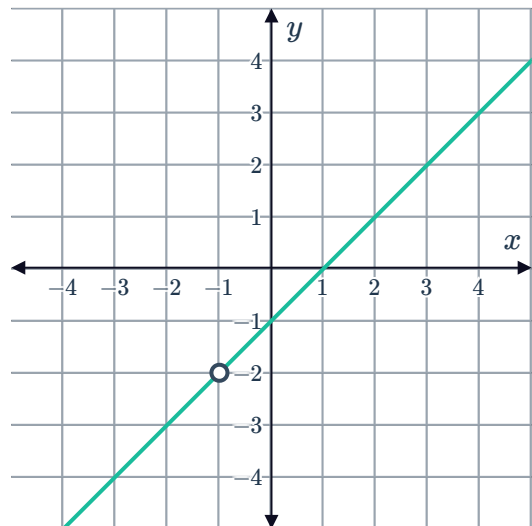
9 Consider the graph of the piecewise function $g(x)$:

- a If we start at -4 and move along $g(x)$ to the right towards $x = -2$, what y -value do we approach?
- b If we start at 0 and move along $g(x)$ to the left towards $x = -2$, what y -value do we approach?
- c Does the $\lim_{x \rightarrow -2} g(x)$ exist? Explain your answer.

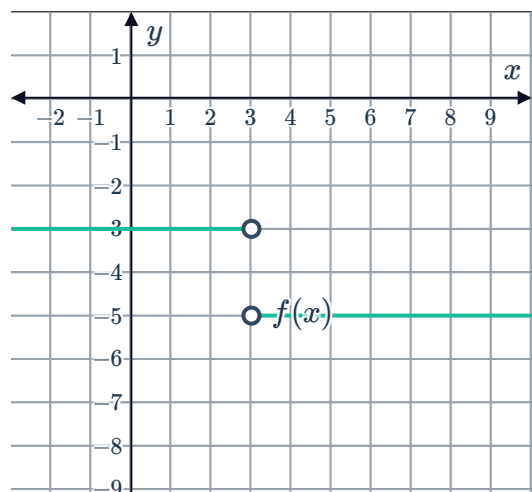


10 Consider the graph of the function $f(x) = \frac{x^2 - 1}{x + 1}$:

- a If we start at $x = -3$ and move along the function to the right towards $x = -1$, what y -value do we approach?
- b If we start at $x = 1$ and move along the function to the left towards $x = -1$, what y -value do we approach?
- c Describe the behaviour of $f(x) = \frac{x^2 - 1}{x + 1}$ as x approaches -1 .
- d Write part (c) using limit notation.

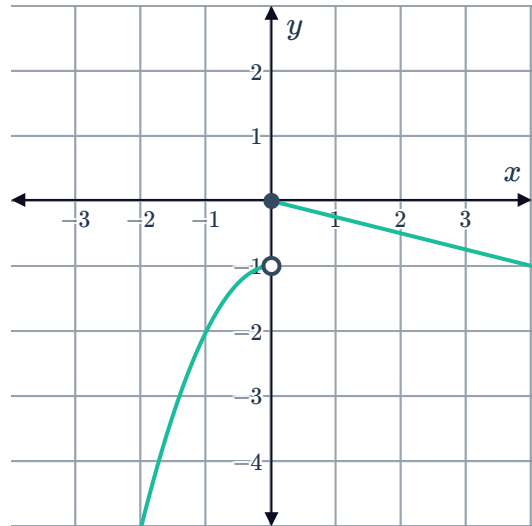


11 Consider the graph of $f(x)$:
Does the limit $\lim_{x \rightarrow 3} f(x)$ exist? Explain your answer.

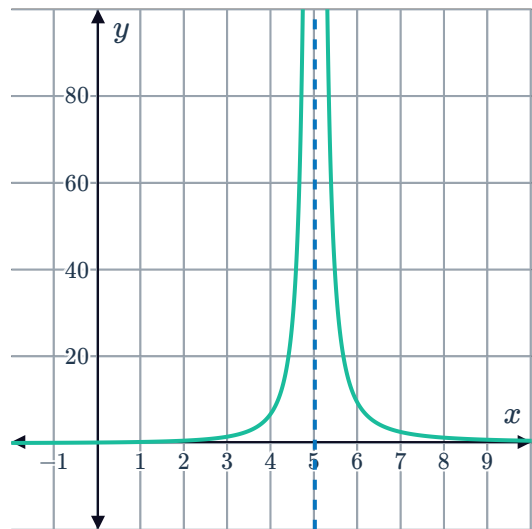




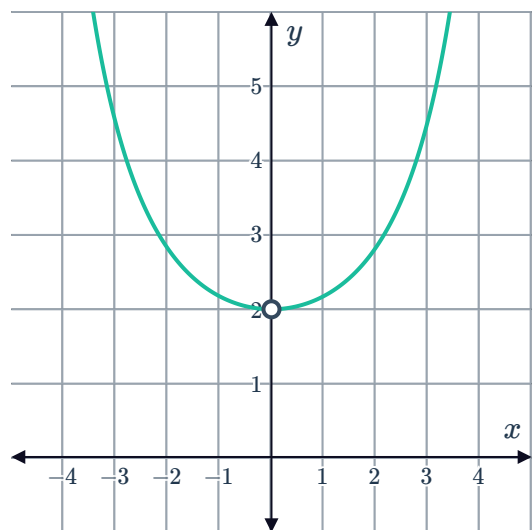
- 12 Consider the graph of $f(x)$:
Does the limit $\lim_{x \rightarrow 0} f(x)$ exist? Explain your answer.



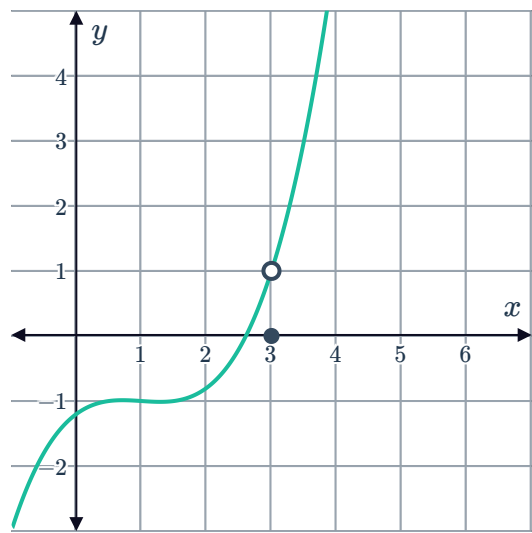
- 13 Consider the graph of $y = \frac{x+3}{(x-5)^2}$:
Does the limit $\lim_{x \rightarrow 5} \left(\frac{x+3}{(x-5)^2} \right)$ exist?
Explain your answer.



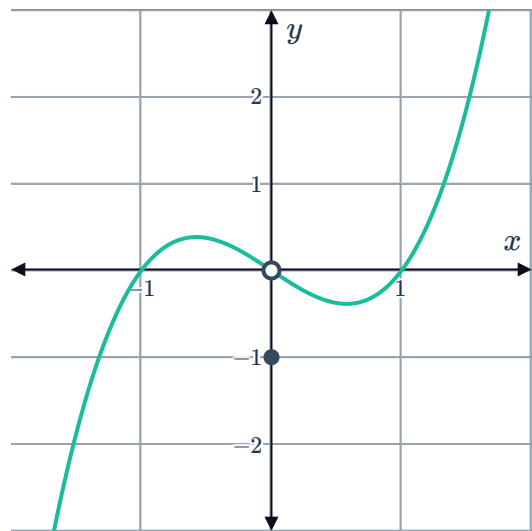
- 14 Consider the graph of $y = \frac{x^2}{1 - \cos x}$:
Does the limit $\lim_{x \rightarrow 0} \left(\frac{x^2}{1 - \cos x} \right)$ exist?
Explain your answer.



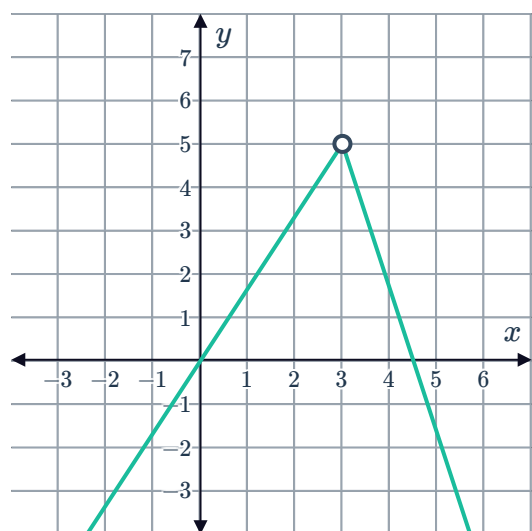
- 15 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \rightarrow 3} f(x)$.



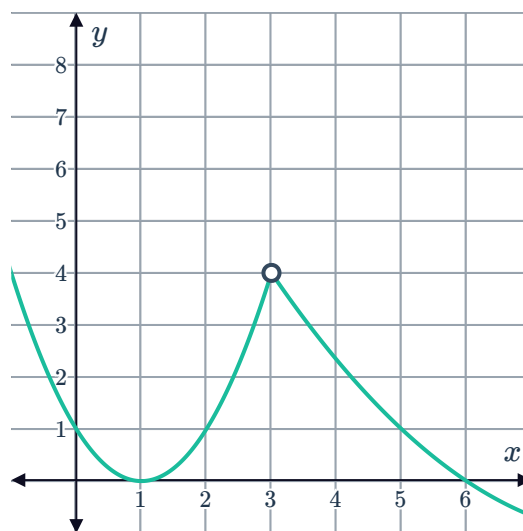
- 16 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \rightarrow 0} f(x)$.



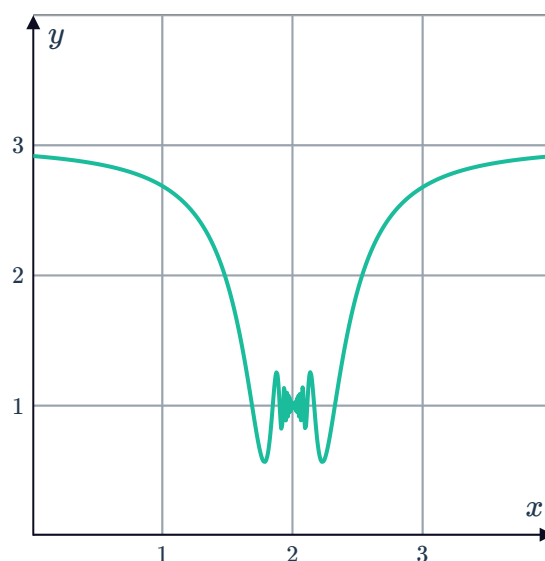
- 17 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \rightarrow 3} f(x)$.



- 18 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \rightarrow 3} f(x)$.



- 19 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \rightarrow 2} f(x)$.



- 20 Consider the graph of the function

$$f(x) = \frac{1}{x+4}:$$

Find the value of:

a $\lim_{x \rightarrow -4^+} \left(\frac{1}{x+4} \right)$

b $\lim_{x \rightarrow -4^-} \left(\frac{1}{x+4} \right)$

